

Question

V_2O_5 and S_2Cl_2 react at a 1:3 ratio at high temperature to produce three substances.

Select the correct option regarding the balanced chemical reaction equation.

- A.** All other options are incorrect
- B.** The product of all coefficients in a chemical reaction equation is 108.
- C.** The product of all coefficients in a chemical reaction equation is 1860.
- D.** The molar ratio of sulfur-containing products in the reaction products is 1:1
- E.** Vanadium-containing reaction product partial hydrolysis will generate vanadium-oxygen double bonds.
- F.** The coefficient of vanadium pentoxide in the reaction equation is 1.

Answer

Correct Answer: **A**

Detailed Explanation

Vanadium pentoxide has strong oxidizing properties and can oxidize +1 valent sulfur to sulfur dioxide at high temperatures. With a reactant ratio of 1:3, vanadium is insufficient to oxidize all sulfur to sulfur dioxide, inevitably leaving some sulfur in a lower valence state, the most reasonable form of which is elemental sulfur. Disulfur dichloride can act as a chlorinating agent, so the products should be vanadium chloride, sulfur dioxide, and elemental sulfur.

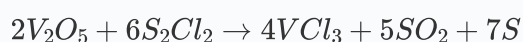
Among vanadium chlorides, VCl_3 and VCl_4 are relatively stable. Since there is an excess of low-valent sulfur present here, the lower valence state VCl_3 is more reasonable.

CHECKPOINT

1 PTS

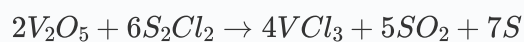
The products are SO_2 , VCl_3 , and S .

The balanced chemical reaction equation is:



CHECKPOINT

2 PTS



The product of all the coefficients in the chemical reaction equation is 1680. Therefore, options B and C are incorrect.

The molar ratio of sulfur-containing products is 5:7. Therefore, option D is incorrect.

Partial hydrolysis of vanadium trichloride cannot generate vanadium-oxygen double bonds. Therefore, option E is incorrect.

CHECKPOINT

1 PTS

Partial hydrolysis of VCl_3 cannot generate vanadium-oxygen double bonds

The coefficient of vanadium pentoxide is 2. Therefore, option F is incorrect.

There is no correct option, so choose option A.